

Course Syllabus for 2110403

1. Course Number: 2110403
2. Course Credit: 3 (3-0-6)
3. Course Title: Introduction to Data Science and Data Engineering
4. Department: Computer Engineering
5. Semester: First Semester
6. Academic Year: 2026
7. Instructor: Prof. Peerapon Vateekul, Ph.D.
Asst. Prof. Natawut Nupairoj, Ph.D.
Assoc. Prof. Veera Muangsin, Ph.D.
8. Condition: -
9. Status: Required
10. Curriculum: Computer Engineering & Digital Technology (CEDT)
11. Degree: B.Eng.
12. Hours/Week: 3-Hour Lecture & Lab
13. Course Description

Data Science is the study of the discovery of knowledge from data. Being a data scientist requires an integrated skill set spanning mathematics, statistics, machine learning, databases, and other branches of computer science, along with a good understanding of the craft of problem solving. Data Engineering is the study of how to engineer or process data, i.e., data cleansing, data storing, etc. There are three main parts in this course:

- Data analysis: Machine Learning techniques
- Data engineering: Data exploration & preparation
- Data visualization: Storytelling via data

14. Course Outline

14.1. Learning Objectives

- Describe what Data Science and Data Engineering are and the skill sets needed
- To be able to explore and understand collected data
- To be able to analyze data by applying traditional machine learning techniques
- To be able to visualize data in relation to spatial and temporal points of view

14.2. Learning Contents

- Sections 51, 52, 53, 54, and 55 on the 3rd floor of the Chula Phat Building 4.
- 2nd Block – Weeks 10 to 18 (October 6 – December 2, 2026)
- Class schedule: Tue 9AM – 12PM, Wed 9AM – 12PM & 1PM – 4PM
 - There are two lecture sessions each week.
 - The Wednesday afternoon session is reserved for practice.
- *** Please bring your laptop to the class ***
- Students need to attend the class on-site at least 80% as a mandatory criterion to “pass” this course.

W	#	DoW	Date	Topics	Professor	Module
10	1	Tue	6-Oct-26	No Class		
10	2	Wed	7-Oct-26	DS1: Introduction, Pandas, Data Prep + AI-Assisted DS2: Data preparation	Aj.Peerapon	DS1, DS2
11	3	Tue	13-Oct-26	No class (H.M. King Bhumibol Adulyadej's Memorial Day)		
11	4	Wed	14-Oct-26	DS3: Decision Tree, Random Forest, Gradient Boosting DS4: Linear Regression, Logistic Regression + kNN, GridSearch	Aj.Peerapon	DS3, DS4
12	5	Tue	20-Oct-26	Self-Study (Online)		
12	6	Wed	21-Oct-26	DS5: Unsupervised Learning DS6: Deep Learning	Aj.Peerapon	DS5, DS6
13	7	Tue	27-Oct-26	Exam Practice on Tue 27 Oct 2026 Morning (7.30AM – 12.30PM)		Extra
13	8	Wed	28-Oct-26	DS7: Advanced topics (Prompt Engineering, RAG, Agentic AI) Tools: LangChain, LangGraph	Aj.Peerapon	DS7
			1-Nov-26	Midterm Exam on Sun 1 Nov 2026 (8.30AM - 11.30AM & 1.00PM - 4.00PM)		
14	9	Tue	3-Nov-26	Data visualization	Aj.Veera	Viz1
14	10	Wed	4-Nov-26	Data storage (Data Lakehouse; Open Table Format) Vector database	Aj.Natawut	DE1
15	11	Tue	10-Nov-26	Web scraping (LLM) + Data ingestion (Kafka)	Aj.Natawut	DE2
15	12	Wed	11-Nov-26	Big data processing (Spark)	Aj.Natawut	DE3
16	13	Tue	17-Nov-26	MLOps: Orchestration (Airflow) and serving (FastAPI, Seldon Core)	Aj.Natawut	DE4
16	14	Wed	18-Nov-26	MCP, AI Integration, Harness Engineering, Evaluation, Guardrails, Observability	Aj.Natawut	DE5
17	15	Tue	24-Nov-26	Viz2: Spatial analysis	Aj.Veera	Viz2
17	16	Wed	25-Nov-26	Viz3: Graph analysis Exam Practice on Wed 25 Nov 2026 Afternoon (12.00PM - 5.00PM)	Aj.Veera	Viz3
18	17	Tue	1-Dec-26	No class		
		Wed	2-Dec-26	Final Exam on Wed 2 Dec 2026 (8.30AM- 11.30AM & 1.00PM - 4.00PM)		

14.3. Method: Lecture and Lab

14.4. Learning Media: PowerPoint presentation, Zoom

14.5. Evaluation

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|----------------|----------------|
| ● Attendance | 5% |
| ● Assignment | 20% |
| ● Project | 15% |
| ● Midterm Exam | 30% (Lab Test) |
| ● Final Exam | 30% (Lab Test) |

15. Reading List

15.1. Required Text: N/A

15.2. Electronic Media or Websites: N/A

16. LMS

16.1. MyCourseVille: <https://mycourseville.com/?q=courseville/course/86514>

16.2. Discord: <https://discord.gg/na8GkXVp2B>

16.3. Github: https://github.com/pvateekul/2110403_DSDE-CEDT_2026s1

16.4. Grader: <https://dsde.nattee.net/>